

Acute effects of psychological relaxation techniques between two physical tasks

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Abstract

The concept of recovery strategies includes various ways to achieve a state of well-being, prevent underrecovery syndromes from occurring and re-establish pre-performance states. A systematic application of individualised relaxation techniques is one of those. Following a counterbalanced cross-over design, 27 sport science students (age 25.22 ± 1.08 years; sports participation 8.08 ± 3.92 h/week) were randomly assigned to series of progressive muscle relaxation, systematic breathing, power nap, yoga, and a control condition. Once a week, over the course of five weeks, their repeated sprint ability was tested. Tests (6 sprints of 4 s each with 20 s breaks between them) were executed on a non-motorised treadmill twice during that day intermitted by 25 min breaks. RM-ANOVA revealed significant interaction effects between the relaxation conditions and the two sprint sessions with regard to average maximum speed over all six sprints, $F(4,96) = 4.06$, $P = 0.004$, $\eta_p^2 = 0.15$. Post-hoc tests indicated that after systematic breathing interventions, $F(1,24) = 5.02$, $P = 0.033$, $\eta_p^2 = 0.18$, participants performed significantly better compared to control sessions. As the focus of this study lied on basic mechanisms of relaxation techniques in sports, this randomised controlled trial provides us with distinct knowledge on their effects, i.e., systematic breathing led to better performances, and therefore, seems to be a suited relaxation method during high-intensity training.

Keywords

Stress; recovery; high-intensity performance; competition; training

Introduction

Stress is ubiquitous in sports. Goals have to be achieved, records to be broken, and rivals or even colleagues outperformed. Getting better than others implies working more, better, and more efficiently; however, the question is, how to accomplish these goals if training regimens are packed and no time is left for additional sessions (Bengtsson, Ekstrand, & Hagglund, 2013; Ekstrand, Walden, & Hagglund, 2004a, 2004b)? Several studies in sports and related disciplines have dealt with a variety of topics concerning the optimisation and further development of training. The priority so far was set on training loads and focus was on the improvement of these aspects. Only recently, based on the realisation that athletes often experience training programmes and competition calendars as overloaded, research has started to address optimised recovery between training units and/or competitions (Ivarsson & Johnson, 2010; Nédélec et al., 2012, 2013).

In 2002, Kellmann outlined a model on the interrelationship between stress and recovery. This so-called scissors model describes the interplay between stress states and recovery demands of an individual and is based on the assumption that one has only limited resources to compensate for (Kellmann, 2002, 2010). Therefore, if stress increases, recovery demands increase consequently. As this happens, individuals have to adapt accordingly. Failing to do so may lead to even more stress, which entails the need for more time for recovery. This situation can be managed until the recovery demands surpass the resource limit or the stress states exceed the stress capacity. However, being high on stress does not necessarily imply negative consequences, as long as recovery demands are kept accordingly. In the long run, however, stress will accumulate and without intervention stress syndromes are much more likely to occur (eg., overtraining, staleness, burnout or even depression; Andersen & Williams, 1999; Bishop, Jones, & Woods, 2008; Budgett, 1998; Kellmann, 2010; Nixdorf, Frank, Hautzinger, & Beckmann, 2013). Relating to the scissors model, the question arose of how accumulating recovery demands can be met. This issue was addressed by Birrer and Morgan (2010), who described a selection of psychological interventions that are the base of world-class performance, such as goal setting, imagery, relaxation and mindfulness. Psychologists and especially sports psychologists try to address the combination of mental and physiological components of this approach (Kellmann, 2002). Following Birrer and Morgan, trained recovery skills are essential for coping with training scope, training intensity,

training years, injury and duration/intensity/continuity of competition impact. A promising opportunity that potentially fits into congested training regimens is the approach of raising recovery quality above quantity. Raising the quality of recovery time between two training stimuli consists of recovering consciously and proactively through specific relaxation techniques.

As elite sports performance is highly demanding both physically and cognitively (Martins, Garganta, Oliveira, & Casanova, 2014; Walsh, 2014), each type of stress should be addressed with a specific recovery method, that is dependent on the type of stress and one's individual capacities (Davidson & Schwartz, 1976; Lehrer, 1996). This matching hypothesis postulates that physical stress should be addressed through somatic-based relaxation, e.g., progressive muscle relaxation (PMR), biofeedback or breathing techniques, and cognitive stress through cognitive-based relaxation, e.g., autogenic training, hypnosis or meditation. However, there is very little research on psychological relaxation techniques in sports contexts, even though they are commonly used (Kudlackova, Eccles, & Dieffenbach, 2013; Martin, Vause, & Schwartzman, 2005). Sport psychologists often advocate the use of relaxation skills to athletes and coaches, even though a lack of data in this area limits our current and past understanding and expertise (Weinberg & Comar, 1994). The application of standardised versions of those methods in clinical settings is integrated routinely and provides an exact and differentiated understanding of relaxation. That includes amongst other effects, a shortening of healing time and/or the reduction of stress (Broadbent et al., 2012), and a positive impact on rehabilitation following sport injury (Schwab Reese, Pittsinger, & Yang, 2012). A review on the use of PMR in clinical trials revealed moderate to large effect sizes on the effectiveness of PMR (Crawford et al., 2013). The use of both cognitive- and somatic-based interventions is frequently mentioned when psychological recovery is the topic of interest in sports (Kudlackova et al., 2013), but to what extent are these methods based on experimental demonstrations of effectiveness and/or efficacy in sport settings? Following Martin and colleague's approach (2005) that focused on psychological interventions with athletes in competition the question is raised: is it worth using them? Clinical research shows that practising these techniques fosters self-control, concentration, an increase in well-being and relaxation (Isa, Moy, Abdul Razack, Zainuddin, & Zainal, 2013; Seers & Carroll, 1998). The relaxation response, elicited through relaxation techniques, manifests itself through short- and

long-term effects in terms of modulation of the central nervous system and reduction of sympathetic activation. Neuro-muscular, cardio-vascular, respiratory and electro-dermal changes are the key areas within these effects (Broadbent et al., 2012; Kellmann & Beckmann, 2014; Schwab Reese et al., 2012).

Especially in clinical contexts, the acute (and long-term) effects of relaxation techniques have been proven to be supportive (Broadbent et al., 2012; Hanton, Wadey, & Mellalieu, 2008; Kanji, 2000; Kanji, White, & Ernst, 2006). In sports, this has only been shown sporadically. A systematic review on relaxation techniques in sports performance contexts (Pelka et al., 2015) found that research is far away from providing valid guidelines for the applied field and that more well-designed trials need to be established. Therefore, the present randomised and controlled study was conducted to investigate whether relaxation techniques affect physical performance acutely in terms of letting body and mind relax while being ready to perform again after a given amount of time. This design, implementing a recovery strategy between two equal sprint protocols, was chosen because of the few studies that tested the effects of relaxation techniques on performance most of them used anaerobic sprinting or swimming tasks as performance indicators (Bar-Eli & Blumenstein, 2004; Bar-Eli, Dreshman, Blumenstein, & Weinstein, 2002; Blumenstein, BarEli, & Tenenbaum, 1995; Donohue et al., 2006). A second reason for choosing the design was of ecological nature. Through the special treadmill and the sprinting performances, we were able to induce fatigue very fast and consistently in all participants (e.g., Hughes, Doherty, Tong, Reilly, & Cable, 2006; Oliver, Williams, & Armstrong, 2006). Moreover, as our participants were sport science students with backgrounds in different sports, the sprinting task was the common feature everyone was familiar with.

In a survey on the usage of psychological-based relaxation in sports, Kudlackova et al. (2013) found that PMR, yoga, breathing and short naps were within the most frequently mentioned strategies. Based on a small number of studies, these strategies have been proven to have the potential to influence physical performance (e.g., Dolbier & Rush, 2012; Donohue et al., 2006; Kim & Tennant, 1993). Exemplary, Donohue et al. (2006) revealed that 20 min sessions of brief single session yoga exercises led to significantly greater improvements in one mile trial run times than 20 min of attention control. Dolbier and Rush (2012) investigated whether 20 min of abbreviated PMR enhanced physiological and psychological functioning in high stress col-

lege students. They found that compared to the control group, the experimental group showed greater increases in mental and physical relaxation as well as lowered cortisol levels. Focusing on breathing, Kim and Tennant (1993) showed that 10 min of a Danjeon breathing exercise enhanced target shooting with an air pistol in a repeated measurement design. Following those outcomes, we expected the yoga exercises to have an enhancing effect on sprinting performance and self-ratings on recovery and well-being. The same was expected for performances and self-ratings after engaging in systematic breathing. PMR has been shown to affect physiological and psychological functioning positively; we therefore assumed that PMR has the same effects on sprinting performance and self-ratings of one's recovery states. Following research by Hayashi, Motoyushi, and Hori (2005) and a review by Lovato and Lack (2010) on the effects of power naps on cognitive functioning, naps that last between 10 and 20 min result in a boost of energy and alertness because of not entering deep sleep states. Therefore, performance improvement is hypothesised; however, most of the relevant research was carried out outside the sport sector.

Taking direct comparison between interventions into account, research by Matsumoto and Smith (2001) has to be considered. Based on the fact that PMR and yoga end with deep relaxation periods (Jacobson, 1938; Yoshihara, Hiramoto, Oka, Kubo, & Sudo, 2014) and systematic breathing requires the attention of a participant throughout the entire session (in which the focus lies on concentrating on one's body function to regulate it and thereby experiencing a relaxing response; Matsumoto & Smith, 2001), they found that responses are different for each relaxation technique as well. In their study, participants completed a relaxation inventory (Smith Relaxation States Inventory; Smith, 2001) after taking part in a PMR or breathing session. Analyses showed that PMR led to a greater increase in the scales *Physical Relaxation* and *Disengagement* while breathing led to higher levels of *Awareness* and perceived *Strength*. Therefore, it was hypothesised that performances after the systematic breathing intervention will be better compared to those after PMR and yoga interventions.

Methods

Participants

Twenty-seven sport science graduate students, 23–32 years old (mean age 25.22 years, $s = 1.08$ years; 8 female), participated in the present study, all with a background in competitive sports. The average body mass index (BMI) was 22.83 kg/ m² ($s = 2.44$) and the average amount of sports participation during a regular week was 8.08 h ($s = 3.92$), 12 participants were mainly involved in individual sports (e.g., track and field, tennis) and 15 in team sports (e.g., football, handball). All of them were fully informed about the content of the study and provided their written consent before participation. None of the participants took part in external relaxation courses throughout the whole experimental phase. Furthermore, none had in-depth knowledge on relaxation techniques (besides having tried out a single technique once before). Three out of an initial sample of 30 participants dropped out due to minor injuries; consequently, 27 were considered for statistical analyses. Ethical clearance for the study was obtained by the ethics committee of the local Faculty of Psychology, following the Declaration of Helsinki.

Design and procedures

The present study was divided in a habituation and intervention sessions. Participants came in for six consecutive weeks, starting with the habituation in week one. During the following five weeks, participants underwent intervention sessions once a week always on the same working day and approximately at the same time (not more than one hour difference). Consequently, a cross-over design with repeated measurements was applied in which the order of intervention conditions and control condition was counterbalanced. The random assignment of the participants to respective intervention orders was based on algorithmic calculation.

The habituation phase was administered to familiarise the participants with the study design, the premises and the material, especially the non-motorised treadmill (NMT, Woodway, Germany), and to collect demographic data. The participants underwent a 15 min familiarisation protocol during which they had to walk/run/sprint at different tempi. This first part was followed by an introduction to the 6 × 4 s sprint protocol.

At the beginning of the intervention sessions, participants were asked to complete the Acute Recovery and Stress Scale (ARSS; Kellmann, Kölling, & Hitzschke, in press) and its short form, the Short Recovery and Stress

Scale (SRSS; Kellmann et al., in press). Then participants were guided to the location of the NMT where they underwent a warm-up protocol and stretching phase before the first 6×4 s sprint test. To guarantee fatigue, another four 4 s sprints were scheduled. At this point, the SRSS was completed again. Thereafter, participants were brought to the relaxation room, which was slightly darkened, had no windows and was the same for all conditions. For the following 25 min, participants underwent one of the five conditions (yoga, PMR, systematic breathing, power nap or the control condition). Following which the SRSS was completed for the third time. Finally, participants had to repeat the sprinting procedure exactly as it was done previously, i.e., warm-up, stretching, and 6×4 s sprints. Before leaving, they had to complete the SRSS once more and evaluate the relaxation condition of the day. Heart rate was monitored throughout the entire sessions.

Performance measures

Performance was assessed on a NMT. The use of the NMT allows for the implementation of self-defined maximum sprints in intermittent exercise, i.e., “all-out pacing” sessions during which participants started each interval as hard as they could and then held on for 4 s (Chia & Lim, 2008; Highton, Lamb, Twist, & Nicholas, 2012; Hughes et al., 2006; Lakomy, 1987). The data output was composed of various variables of which two were selected for the present study. These are the average maximum speed (m/s) and the maximum power output (watts). Therefore, the differences in average maximum speed and average maximum power between two 6×4 s sprint sessions (with 20 s breaks between sprints) were integrated as the key objective outcome measures.

Physiological measures

Heart rates of the participants were tracked throughout the entire test sessions using Polar RS800CX (Polar, Finland) devices. The resting heart rate (5 min), the heart rate during the first and second sprint session (2.20 min each) and the heart rate during the relaxation period were included in the analysis.

Psychological measures

The different instruments used during the study were the original German versions of the ARSS and SRSS. Both were integrated to represent the recovery/stress states of the participants, before and during the experiments. In addition, basic demographic and anthropometric data were gathered prior to the experimental phase. The demographics consisted of questions on participants' age, hours spent on sports and the type of sport. A manipulation check was administered after every session to gain insights on participants' evaluations of the interventions.

Acute Recovery and Stress Scale

The ARSS consists of 32 adjectives, which are clustered into eight scales, assessing recovery states on four scales (*Physical Performance Capability*, *Mental Performance Capability*, *Emotional Balance* and *Overall Recovery*) and stress states on the other four (*Muscular Stress*, *Lack of Activation*, *Negative Emotional State* and *Overall Stress*). Example items include *recovered*, *balanced* and *receptive*. Answers had to be indicated on a 7-point Likert scale (0–6) with end points, “does not apply at all” and “fully applies”. Kellmann et al. (in press) reported satisfactory internal consistencies of at least $\alpha = .84$ for all eight scales. This questionnaire was administered to evaluate participants' states before the experiment.

Short Recovery and Stress Scale

During the experimental phase, recovery/stress states were assessed using the SRSS. This short form is composed of eight items, derived from the originally 32 items of the ARSS. These items are similar to the eight scales of the ARSS, accordingly categorised into four recovery- and four stress-related items. Answers had to be indicated on the same 7-point Likert scale as the ARSS. Internal consistencies for all scales of the SRSS were found to be satisfactory ($\alpha > .75$).

Manipulation Check

The manipulation check addressed how participants experienced the recovery period on two items. On the one hand, there was the self-rated efficacy (“How effective did you experience the relaxation technique”) and on the other the appreciation of the intervention method (“How did you like the relaxation technique”).

Answers had to be given on 7- point Likert scales, ranging from “not at all” to “very much”. After participants had undergone all five conditions, they had to rank them on both, self-rated efficacy and appreciation again.

Interventions

The relaxation techniques introduced during the recovery phases were PMR, relaxational yoga exercises and systematic breathing. The PMR session consisted of a standardised 20 min protocol of tensing and relaxing specific muscle groups (instructions according to Kellmann & Beckmann, 2014). The yoga protocol was built up in cooperation with an experienced and licenced yoga teacher and included a 5 min introductory phase, followed by three rounds of sun salutations (5 min), 2 rounds of triangle A (4 min) and 2 min of baddha konasana and 5 min of savasana (Yoshihara et al., 2014). The foundations of the breathing exercise based on that the exhalation had to be twice as long as the inhalation period, leading to 3 s/6 s or 4 s/8 s rhythms. As a fourth recovery method, power naps (introduction and 20 min time for a nap) were applied. These four strategies were complemented by a control condition in which the participants had the possibility to read/go through a collection of comics. All strategies were introduced and executed within 25 min (actual practice phase of the breathing regulation was 10 min, instructions to find their appropriate breathing rhythm took 10 min extra) with the same researcher in each of the 135 sessions. Language of instruction was German, as all participants' were native speakers (except for one who, however, was fluent in German). The experimenter was blind to all hypotheses.

Data analyses

IBM SPSS Statistics 22 was used to analyse data. To test whether relaxation techniques had an influence on performance, Repeated Measures Analyses of Variance (5×2 RMANOVA) were executed. Twenty-seven participants underwent all five conditions and were included in the analyses. To secure an adequate analysis and to check for prerequisites that could have influenced behaviour and performance throughout the experimental sessions, the initial states of recovery-stress balances, heart rate and sprint performances were compared. Following Cohen's (1973) argumentation on manipulated variables, the effect sizes were reported using partial eta squared (η^2_P). Values can be evaluated as small (0–0.029), medium (0.03–0.159) and large (0.16–0.35) (Cohen, 1992).

Results

To examine whether initial differences in emotional and physical recovery states were applicable, results of the anamnesis questionnaires of each experimental day were compared as well as initial running performances (see Table 1). RM-ANOVA revealed no significant differences in the ARSS, $F(4,96) = 1.42$, $P = 0.234$. Two RM-ANOVAs showed no differences between conditions in first session running performances and resting heart rates either. Neither analysis revealed differences in preperformance states. Therefore, further calculations did not have to take pre-performance differences into account. Additionally, an acceptable drop-out rate of three out of 30 participants can be reported. All other participants underwent the habituation and five intervention sessions. Both age and individual fitness levels in terms of BMI scores did not affect results.

Performance measures

To examine whether there were differences between the experimental conditions regarding the performance measures, a RM-ANOVA was executed. At first, the average maximum speed on the NMT over the six sprints of a session was evaluated. Analyses yielded no main effect for treatments, i.e., differences between the experimental conditions, $F(4,96) = 0.15$, $P = 0.963$, $\eta_p^2 = 0.01$. A main effect for time, i.e., the overall comparison between pre- and post-test session, was not found either, $F(1,24) = 4.06$, $P = 0.061$, $\eta_p^2 = 0.15$. However, the interaction between treatments and measurement points was significant, $F(4,96) = 4.06$, $P = 0.004$, $\eta_p^2 = 0.15$. Post-hoc calculations with Bonferroni correction revealed significant differences between breathing regulation and the control condition, $F(1,24) = 5.10$, $P = 0.033$, $\eta_p^2 = 0.18$, as well as between power naps and the control condition, $F(1,25) = 4.83$, $P = 0.037$, $\eta_p^2 = 0.16$. Participants increased their average maximum speed from pre- to post-relaxation intervention only in the systematic breathing regulation, $t(25) = -2.13$, $P = 0.043$, and the power naps conditions, $t(26) = -2.993$, $P = 0.006$. All other post-hoc comparisons between the experimental groups and the control group and pre-to-post-test changes in the two other experimental groups were non-significant. Performance differences with regard to the average maximum power output (in watts) were assessed analogously with RM-ANOVA. No main effects were found for treatment differences, $F(4,96) = 0.40$, $P = 0.600$, $\eta_p^2 = 0.02$, and differences between measurement points overall,

$F(1,24) = 1.05$, $P = 0.316$, $\eta_p^2 = 0.04$. Again, the interaction between treatment and measurement points was found to be significant, $F(4,96) = 4.25$, $P = .003$, $\eta_p^2 = 0.15$. Post-hoc tests showed no differences between single conditions. Differences between systematic breathing and the control condition were the only comparison close to significance, $F(1,24) = 3.66$, $P = 0.070$, $\eta_p^2 = 0.13$. None of the interventions led to post-relaxation improvements in the average maximum power output.

Table 1 – The average maximum speed (m/s) in pre- and post-relaxation sprints.

Treatments	Pre-relaxation	Post-relaxation
Control	5.19 ± .65	5.19 ± .64
Breathing	5.17 ± .65	5.21 ± .61
Yoga	5.19 ± .65	5.19 ± .64
Power nap	5.19 ± .65	5.23 ± .63
PMR	5.19 ± .65	5.19 ± .64

Physiological measures

Comparing mean heart rates from the first sprint session, the relaxation phase and the second sprint session revealed a significant main effect for measurement points, $F(2,42) = 870.82$, $P < 0.001$, $\eta_p^2 = 0.98$, and a significant interaction effect between treatments and measurement points, $F(8,168) = 5.94$, $P < 0.001$, $\eta_p^2 = 0.22$. There was no significant difference between treatments, $F(4,84) = 0.28$, $P = 0.585$, $\eta_p^2 = 0.01$. The yoga intervention led to a significantly higher heart rate than the other interventions during the relaxation period, control condition versus yoga, $t(23) = -5.604$, $P = 0.001$; systematic breathing versus yoga, $t(23) = -5.435$; $P < 0.001$; power nap versus yoga, $t(22) = 6.298$, $P < 0.001$; PMR versus yoga, $t(23) = -5.885$, $P < 0.001$; nonetheless, no differences were found during the two sprint sessions. For exact numbers, see Table 2.

Table 2 – The average maximum speed (m/s) in pre- and post-relaxation sprints.

Treatments	Pre-relaxation sprint	During relaxation	Post-relaxation sprint
Control	162.36 ± 12.41	87.79 ± 12.84	156.16 ± 19.78
Breathing	160.36 ± 12.49	86.56 ± 11.90	161.24 ± 14.13
Yoga	159.54 ± 15.01	96.56 ± 9.71	154.73 ± 15.30
Power nap	161.56 ± 12.79	85.86 ± 11.50	160.03 ± 13.56
PMR	161.30 ± 10.20	86.60 ± 11.12	156.64 ± 13.03

Psychological measures

Subjective overall recovery scores measured with the SRSS (see Figure 1) changed significantly throughout the course of the experimental phases, $F(3,75) = 44.57$, $P < 0.001$, $\eta_p^2 = 0.64$. However, there appeared to be no differences between treatments, $F(4,100) = 2.65$, $P = 0.053$, $\eta_p^2 = 0.10$, and interactions between treatments and measurement points, $F(12,300) = 1.13$, $P = 0.332$, $\eta_p^2 = 0.04$.

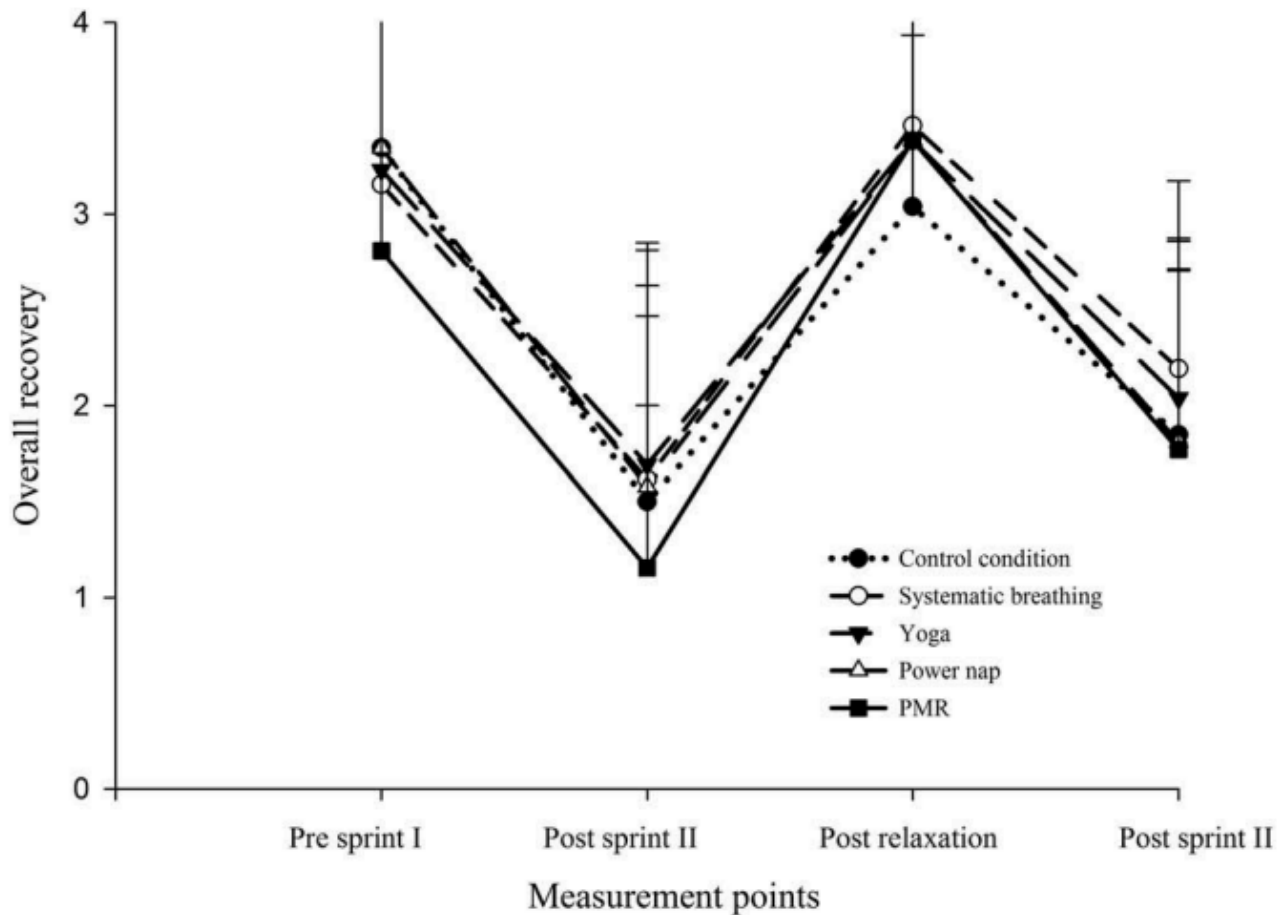


Figure 1 – Study design and course of testing day.

Note. The vertical scale is downsized from 0–6 (the original item scaling) to 0–4 for visual clarity

Similar results were found for the overall stress scores [measurement effect: $F(3,75) = 23.03$, $P < 0.001$, $\eta_p^2 = 0.48$; treatment effect: $F(4,100) = 1.99$, $P = 0.102$, $\eta_p^2 = 0.07$; interaction effect: $F(12,300) = 1.79$, $P = 0.085$, $\eta_p^2 = 0.07$]. The SRSS scales showed that participants experienced the sprint sessions as more stressful and the relaxation sessions as more relaxing than their initial states. The manipulation check did not reveal significant differences in the self-rated efficacy and appreciation of the five recovery strategies (Figure 2).

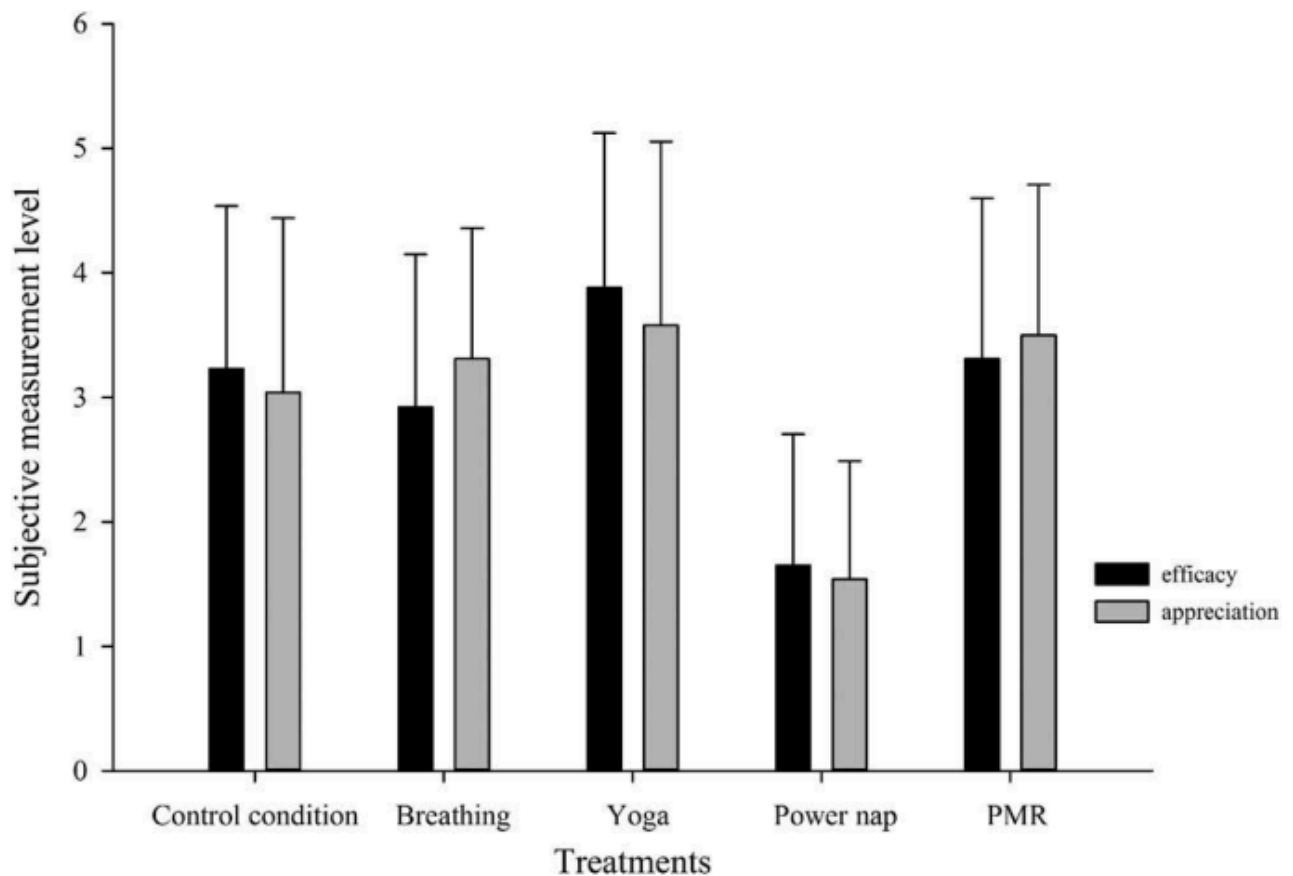


Figure 2 – Subjective description of the relaxation techniques.

Discussion

Given the paucity of research regarding relaxation techniques in sports, this approach of comparing several techniques within one experimental design is even more uncommon. A few studies examined whether relaxation strategies have an influence on subsequent performance but most of the designs contained an experimental and a control condition only (amongst others, Bar-Eli & Blumenstein, 2004; Bar-Eli et al., 2002). Consequently, it could not have been ruled out that other techniques would have led to better outcomes in those particular situations. As the aim of the present study was to examine whether relaxation techniques affect physical performance as well, analyses compared the results of pre- and postrecovery sprint performances taking four different recovery strategies into account.

Performance measures

Analysing performances in terms of changes between sprinting outcomes revealed that the use of breathing regulation and a power nap led to improved running speed compared to the control condition and PMR. Thus, 25 min periods of these two strategies have the potential to actively improve performance. Surprisingly, no long-term practice in these techniques was necessary to affect sprint performances. Taking this issue into account, it has to be mentioned that regulating effects of PMR and yoga are strongly related to practice effects (Kellmann & Beckmann, 2014), even though research found significant effects after single session versions of abbreviated PMR (Carlson & Hoyle, 1993; Dolbier & Rush, 2012; Pawlow & Jones, 2005). That could have been one reason why PMR and yoga did not affect performance outcomes in this setting. However, the above-mentioned studies relied on self-reports and monitoring rather than on objective performance variables; for example, Pawlow and Jones (2005) found that 20 min of abbreviated PMR resulted in significantly reduced self-ratings of perceived stress and state anxiety as well as significantly decreased heart rate compared to a quiet sitting control group. Nevertheless, focusing on the different recovery methods could provide reasons for improvement or stagnation of sprinting performance after the various conditions.

As participants knew that after the relaxation phase they had to undergo the next sprint session, they most likely expected physical strain/stress (Ursin & Eriksen, 2004). This expectation is naturally connected to a short self-assessment, if one is ready to respond to the stimulus. During power nap interventions, activation levels probably decrease. Therefore, participants put in extra effort to wake themselves up because they felt too tired to run after approximately 15–20 min of sleep. In that special case, the stress response would have been a desirable alarm that enables the body to perform immediately, through the mobilisation of physiological resources (Ursin & Eriksen, 2004). Additionally, as mentioned earlier, 2–3 min after waking up, activation and energy levels rise again (Lovato & Lack, 2010). Taking the other performance enhancing intervention into account (i.e., breathing), it has to be stated that activation levels stayed relatively high during the intervention because a consistent conscious engagement towards the breathing rhythm was demanded (Matsumoto & Smith, 2001). Thus, participants were prepared for the next sprint session. On the contrary, during

PMR and yoga, participants followed instructions which demand lower cognitive involvement (Lehrer, 1996) and pro-actively prepare relaxation. Therefore, these techniques probably led to higher physical relaxation which is consistent with Beckmann and Elbe's (2015) advice to not implement relaxation techniques such as PMR and autogenic training before competition. Research by Amihai and Kozhevnikov (2014) points in a similar direction. They found that variations of meditation techniques can lead to different results and are even based on different neurophysiological mechanisms. On the one hand, phasic alertness, a significant temporary boost in the capacity to respond to stimuli and on the other tonic alertness, a state of constant vigilance. While tonic alertness can happen concurrently with relaxation, phasic alertness is a result of the activity of the sympathetic system (Amihai & Kozhevnikov, 2014). With regard to the present setting, those differences could have led to an advantage in mental readiness prior to the second sprint protocol for both systematic breathing and power nap.

The effects sizes found in the present study resemble those medium-to-large effect sizes that were reported in similar studies (Bar-Eli & Blumenstein, 2004; Bar-Eli et al., 2002), taking into account that only very few reported values at all. As our aim was to see whether the mechanisms of psychological relaxation strategies generally affect subsequent performance in physical tasks, we conducted the present study with semi-professional/ recreational athletes. Taking both purpose and target population into consideration, next steps should include the continuation of basic research on mechanisms on one hand and taking findings into professional sport practice on the other.

Physiological measures

Measures of the cardio-vascular system, i.e., changes in heart rate, showed that all interventions did affect the participants' physiological state during experimental sessions (see Table 2) but no differences between interventions were found. Therefore, next steps should include the thorough examination of further mechanisms involved in relaxation, such as a tonus decrease in skeletal muscles, the downward regulation of reflexes (neuro-muscular changes), changes in respiration (e.g., decreases in breathing rate, more balanced breathing cycles) or electro-dermal activity (e.g., a decrease in skin conductance responses).

Psychological measures

As a subjective measure over the course of the intervention sessions, the SRSS (Kellmann et al., in press) was applied. Though analysis did not show differences between the five conditions, it revealed that all recovery strategies and the physical stressor (sprint protocols) had an influence on participants' perceptions and evaluations of their personal recovery and stress states (see Figure 2). Additionally, the manipulation check confirms that participants perceived the conditions as rather relaxing and supportive but not significantly different from each other. However, while there is no statistically relevant difference in the self-reported assessment of efficacy and appreciation of the recovery strategies, a clear tendency is apparent. As Figure 2 shows, power nap has the lowest values for both efficacy and appreciation after taking all experimental conditions into account. It appears to be that all strategies that include pro-active and conscious action taking (PMR, systematic breathing and yoga) were rated as more effective and likeable than "just" laying down (power nap). In the end, these perceptions did not influence performance outcomes.

Limitations

Unfortunately, we did not obtain measures of sympathetic and parasympathetic activation which could have brought us more insight into what happened during the interventions and which physiological mechanisms lay behind the performance developments. Taking the pre-experiment states of participants into account, no initial fitness test was administered. Additionally, no extra tests were implemented to check whether the physical stress was high enough; however, during pre-tests, countermovement jumps had been implemented in a very similar pre- and post-relaxation sprint design. Analysis of those jumps revealed that jump heights were significantly lower after sprint sessions compared to a baseline. Comparing baseline measurements to post-relaxation jumps resulted in no difference. Another concern was whether participants slept, how long they slept and if they slept with or without short awakenings. Regarding this issue, we had to rely on participants' reports.

Conclusions

Following the results of the study, systematic breathing and short power naps can be recommended during performance breaks that allow for 25 min recovery sessions. On the contrary, yoga and PMR are probably not suited for these situations. The hypotheses that these two strategies need more practice than the other two to fully unfold their potential and that they are probably more appropriate in situations with longer “wake-up” phases or even situations without a second performance on the same day need to be examined in further research designs. All in all, recovery in terms of relaxation remains highly individual and needs to be adjusted per athlete (Kellmann, 2010); however, it seems to be that simpler methods such as power naps and systematic breathing have an effect on all participants.

At this point, the importance of implementing evidence-based practice in this area needs to be addressed. Randomised controlled trials are the gold standard in evaluating health care interventions. Nevertheless, if trials lack methodological rigour, results can be biased (Juni, Altman, & Egger, 2001). Therefore, we designed the study with regard to standardised guidelines and reported methods and evaluations as clearly as possible (Furlan, Penninck, Bombardier, & van Tulder, 2009). Especially when mechanisms are the topic of interest, as they were in this study design, this methodological approach seems to be appropriate.

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